Block 4: Introduction to continuum mechanics

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Contents

Chapter 1

Tensor Fundamentals

Section References

The references used in this chapter are:

- Belytschko chapter 3.1 [1]
- Crisfield chapter 1.1-1.3 [2]

This chapter will cover the vector and tensor essentials to understand tensor algebra applied to solid mechanics. Vectors and matrix are essentially containers for scalars. Scalar are ordered in a certain fashion and some operation can be done with them. Tensors are an extension to vector and matrix.

Example 1.0.1: First test with example

This is an example

Key Concept 1.0.1: Einstein notation

This is a concept

1.1 Index Notation & Einstein Summation

1.2 Stress Tensor Derivation

1.3 Strain Tensor Derivation

1.4 Problem sets

1. Einstein Summation
2. Displacement Field
3. Stress Tensor Components
4. Traction Calculation

Bibliography

[1] Ted Belytschko, W. K. Liu, B. Moran, and Khalil I. Elkhodary. *Nonlinear finite elements for continua and structures.* Hoboken, New Jersey : Wiley, 2014.

[2] Joris J. C. Remmers René de Borst, Mike A. Crisfield and Clements V. Verhoosel. *Non-Linear Finite Element Analysis of Solids and Structures.* John Wiley and Sons, Ltd, 2012.